MEASUREMENTS AND EVALUATION

A Full Lecture for Teachers

Faculty of Education | Teacher Professional Development Series
Lecture Duration: 3 Hours | Level: Pre-service and In-service Teachers

Lecture Overview

This lecture provides teachers with a comprehensive, practical understanding of measurement and evaluation in education. Covering foundational theory, test design, statistical interpretation, and authentic assessment strategies, it equips teachers to assess learner performance with validity, fairness, and professional confidence.

LECTURE INFORMATION

Item	Details
Lecture Title	Measurements and Evaluation
Target Audience	Pre-service and in-service teachers (primary to tertiary level)
Duration	3 Hours (may be split into two 90-minute sessions)
Mode	Lecture with discussions, worked examples, and group activities
Prerequisite	Basic understanding of curriculum and classroom instruction
Key Outcome	Teachers will design and use valid, reliable, and fair assessments

LECTURE OBJECTIVES

By the end of this lecture, teachers should be able to:

1. Accurately define and distinguish the terms measurement, assessment, evaluation, and testing.
2. Explain the purposes and principles of sound educational assessment.
3. Identify and describe the major types of assessment instruments and their appropriate uses.
4. Construct a Table of Specifications and use it to design balanced tests.
5. Write high-quality objective and essay test items across Bloom's cognitive levels.
6. Score, grade, and interpret student assessment results with confidence.
7. Apply basic descriptive statistics to classroom data.
8. Design authentic assessment tasks including rubrics, portfolios, and performance tasks.
9. Reflect on ethical issues in assessment including fairness, validity, and confidentiality.

PART ONE: CONCEPTUAL FOUNDATIONS

LECTURE ONE: UNDERSTANDING MEASUREMENT AND EVALUATION

1. Introduction — Why Assessment Matters

Imagine teaching a class for an entire term without ever checking whether your students understood a single lesson. You delivered every topic, gave every example, and answered every question — but you never found out what they actually learned. This is the problem that measurement and evaluation solve.

Assessment is not a bureaucratic obligation or an end-of-term ritual. It is the core feedback mechanism that makes teaching and learning effective. As a teacher, the data you gather from assessments tells you who has mastered the material, who needs more support, which topics need to be re-taught, and whether your own instructional methods are working.

Reflection for Teachers

"Assessment is today's means of understanding how to modify tomorrow's instruction." — Carol Ann Tomlinson

Think about this: When did you last use assessment data to change what you taught or how you taught it? That moment — where data informs instruction — is what good assessment is for.

2. Defining Key Terms

One of the most common sources of confusion in education is the interchangeable use of terms that have distinct meanings. Every teacher must be able to define and distinguish these four foundational concepts:

2.1 Measurement

Measurement is the process of assigning numbers or scores to learner attributes according to established rules. It is quantitative and objective.

Classroom Example

A student answers 18 out of 25 questions correctly on a quiz. The act of counting those correct answers and recording '18/25' is measurement. It tells us the quantity of correct responses — nothing more, nothing less.

Measurement answers: 'How much?' or 'How many?'

2.2 Assessment

Assessment is a broader, ongoing process of collecting evidence about student learning through multiple means — tests, observations, projects, questioning, portfolios — for the purpose of understanding and improving learning.

Assessment is both formative (happening during learning) and summative (happening after learning). It encompasses not just scores but also teacher observations, student self-reflection, and qualitative feedback.

Classroom Example

Throughout a unit on fractions, a teacher uses exit tickets, observes group work, checks homework, asks probing questions, and gives a quiz. All of these together constitute the teacher's assessment of students' understanding of fractions.

Assessment answers: 'What do students know, understand, and can do?'

2.3 Evaluation

Evaluation is a value judgment — a professional decision about the quality, worth, or merit of a student's performance, a curriculum, or a teaching programme. It integrates measurement data with professional judgment to reach a conclusion.

Classroom Example

After reviewing a student's test scores, class participation, project work, and attitude over a term, the teacher concludes: 'This student has made commendable progress but still struggles with analytical reasoning.' That judgment is evaluation.

Evaluation answers: 'How good is it?' or 'What is the value of what has been learned?'

2.4 Testing

Testing is a specific, formal procedure using standardized instruments (called tests) to obtain data about learner performance under controlled conditions. A test is one tool within the broader assessment process.

Key Distinction

All tests are assessments, but not all assessments are tests.

All assessments involve measurement, but measurement alone is not assessment.

Evaluation requires data (from measurement and assessment) plus professional judgment.

Think of it as nested circles: Measurement is the innermost, followed by Testing, then Assessment, and Evaluation as the outermost process.

3. The Relationship Between Teaching, Learning, and Assessment

Assessment is not separate from teaching — it is woven through every stage of the instructional process:

Stage	Role of Assessment
Before Teaching (Pre-Assessment)	Diagnose prior knowledge, misconceptions, and readiness. Informs lesson planning.
During Teaching (Formative)	Monitor ongoing understanding. Provide feedback. Adjust instruction in real time.
After Teaching (Summative)	Measure total learning achieved. Assign grades. Report to parents and administrators.
After Reporting (Evaluative)	Reflect on teaching effectiveness. Identify curriculum gaps. Improve future practice.

This cycle is known as the Assessment-Instruction Loop. Effective teachers move continuously through this loop, using evidence to drive every instructional decision.

4. Purposes of Educational Measurement and Evaluation

Assessment serves multiple purposes that go beyond assigning grades. Every teacher should be able to articulate the specific purpose of each assessment they design and administer:

- Instructional Purpose: Provide actionable feedback to both teachers and learners. Formative assessments serve this purpose primarily.
- Diagnostic Purpose: Identify specific learning difficulties, knowledge gaps, or misconceptions before or during instruction so targeted interventions can be designed.
- Placement Purpose: Determine the appropriate instructional level, programme, or class for a student (e.g., remedial grouping, gifted stream placement).
- Certification Purpose: Confirm mastery of content or skills for academic promotion, graduation, or professional qualification (e.g., WAEC, NECO, JAMB).
- Motivational Purpose: Stimulate student engagement, effort, and goal-setting. Well-designed assessments challenge and inspire learners.
- Programme Evaluation: Assess the effectiveness of a curriculum, instructional method, or school programme. Data informs policy and resource allocation.
- Accountability Purpose: Provide evidence to parents, administrators, and the public that educational standards are being met.

Classroom Activity 1 (10 minutes)

In pairs or small groups, think about three assessments you use regularly in your classroom.

For each one, identify: (a) What type is it? (b) What is its PRIMARY purpose? (c) Does the way you design and use it actually serve that purpose?

Share one insight with the class.

5. Principles of Sound Assessment

Not all assessments are equally good. A professionally designed assessment must meet several fundamental criteria. These principles act as a quality checklist for every assessment instrument you create or use:

5.1 Validity

Validity is the most important property of any assessment. A valid test measures what it is supposed to measure — and nothing else. There are several types of validity:

Type of Validity	What It Means and How to Achieve It
Content Validity	The test adequately covers the full range of learning objectives taught. Achieved by using a Table of Specifications (TOS).
Criterion Validity	Test scores accurately predict or correlate with an external standard (e.g., future performance, another accepted

Type of Validity	What It Means and How to Achieve It
	measure).
Construct Validity	The test genuinely measures the underlying construct it claims to measure (e.g., critical thinking, reading comprehension).
Face Validity	The test appears relevant and reasonable to students and other stakeholders — even if not technically rigorous.

Common Validity Mistake

A teacher teaches critical thinking skills throughout the term but only tests factual recall at the end. The test lacks construct validity — it does not measure what was actually taught and valued.

Always align your test to your learning objectives. If you taught analysis, test analysis.

5.2 Reliability

A reliable test produces consistent results across different occasions, different markers, and different versions of the test. Reliability is a necessary (but not sufficient) condition for validity — a test can be reliable without being valid, but cannot be valid without being reliable.

- Test-Retest Reliability: Administer the same test twice; scores should be consistent.
- Inter-Rater Reliability: Two markers should assign similar scores to the same response. Critical for essay and performance assessments.
- Internal Consistency: Items within the test should measure the same construct consistently (assessed with Cronbach's Alpha).

Classroom Example

Two teachers mark the same student essay on climate change. Teacher A gives 72/100; Teacher B gives 48/100. This wide discrepancy indicates low inter-rater reliability. The solution is a detailed analytical rubric with clear level descriptors that both markers apply.

5.3 Objectivity

Objectivity means freedom from scorer bias. The same answer should receive the same score regardless of who marks it, when it is marked, or how well the teacher likes the student. Objectivity is highest in well-designed objective tests and can be improved in essay tests through structured rubrics and blind marking.

5.4 Comprehensiveness

A comprehensive test samples broadly from the domain of learning objectives. It should not focus narrowly on a small subset of topics or cognitive levels. The Table of Specifications is the primary tool for ensuring comprehensiveness.

5.5 Practicability

An assessment must be feasible within the real constraints of teaching: available time, budget, marking workload, and classroom resources. An instrument that is theoretically perfect but impossible to administer is not useful. Assessment design always involves trade-offs between ideal measurement and practical constraints.

5.6 Fairness and Freedom from Bias

A fair assessment does not systematically disadvantage any group of students based on gender, ethnicity, language background, disability, or socioeconomic status. Teachers must review items for cultural bias, language complexity, and stereotype threat.

Teacher's Checklist: Principles of Sound Assessment

Validity: Does this test measure the learning objectives I actually taught?

Reliability: Would this test produce consistent results if marked by another teacher or administered again?

Objectivity: Are the scoring criteria clear enough to eliminate subjective bias?

Comprehensiveness: Does the test cover the full range of topics and cognitive levels in my TOS?

Practicability: Can I realistically administer, collect, mark, and return this assessment on schedule?

Fairness: Does any item disadvantage students based on background or identity rather than learning?

PART TWO: TYPES AND DESIGN OF ASSESSMENTS

LECTURE TWO: TYPES OF ASSESSMENT AND BLOOM'S TAXONOMY

6. Types of Assessment

6.1 By Timing and Function

Assessment Type	Description and Classroom Use
Diagnostic Assessment	Administered before instruction begins. Identifies prior knowledge, misconceptions, and readiness. Examples: pre-tests, concept maps, KWL charts, question-and-answer warm-ups. Informs lesson planning.
Formative Assessment	Ongoing during the unit or lesson. Monitors progress and provides real-time feedback to both teacher and learner. Examples: exit tickets, short quizzes, observation, questioning, peer feedback, mini whiteboards. Guides instructional adjustments.
Summative Assessment	Administered at the end of an instructional period (unit, term, year). Evaluates overall achievement and assigns grades. Examples: end-of-term exams, final projects, standardized tests.
Placement Assessment	Determines the most appropriate instructional group, level, or programme for a student. Examples: reading level tests, entrance examinations, skills surveys.

Formative vs Summative: A Crucial Distinction

The purpose determines the type — not the format. A quiz CAN be formative (if results guide reteaching) or summative (if used solely for grading).

Black and Wiliam (1998) found in their landmark review that strengthening formative assessment produces some of the largest gains in student achievement ever documented in educational research.

Ask yourself after every assessment: 'What will I do differently because of this data?'

6.2 By Reference Point

Type	Description
Norm-Referenced (NRT)	Compares student performance to a norm (reference) group — usually a national or regional sample. Results are reported as percentile ranks or standard scores. Used for: selection, ranking, national comparisons. Examples: JAMB UTME, SAT, IQ tests.
Criterion-Referenced (CRT)	Measures student performance against a fixed, pre-determined standard — regardless of how other students performed. Used for: mastery learning, certification, competency-based assessment. Examples: WAEC, driving test, teacher licensure.
Ipsative Assessment	Compares a student's current performance to their own previous performance. Focuses on individual growth and progress rather than comparison to others. Highly motivating for struggling learners.

NRT vs CRT in Practice

NRT scenario: A student scores at the 85th percentile on a national mathematics test — they performed better than 85% of students in the norm group. This tells us about their standing relative to peers.

CRT scenario: A student must score at least 50% to pass WAEC Chemistry. Their score is compared to the standard, not to other students. A score of 55% is a pass regardless of whether every other student scored 90%.

7. Bloom's Taxonomy and Its Assessment Implications

Benjamin Bloom and his colleagues (1956) classified educational learning objectives into hierarchical levels, later revised by Anderson and Krathwohl (2001). Every teacher must understand and apply this taxonomy when writing learning objectives and designing assessment tasks — because it is impossible to assess what you have not defined.

7.1 The Cognitive Domain (Revised Bloom's Taxonomy)

Level	Description	Assessment Examples
1. Remember	Recall facts, definitions, sequences, and basic concepts from memory.	List the planets in the solar system. Name the four blood types. State

Level	Description	Assessment Examples
		the formula for photosynthesis.
2. Understand	Explain ideas, interpret meaning, classify, summarize, and describe in own words.	Explain the water cycle in your own words. Summarize the causes of the Nigerian Civil War.
3. Apply	Use knowledge and procedures in new or familiar situations.	Calculate the area of a trapezium. Solve this quadratic equation. Apply Newton's second law to this problem.
4. Analyze	Break down information into components, draw connections, distinguish cause and effect, identify relationships.	Compare the themes in two poems. Distinguish between ionic and covalent bonds. Analyze the causes of inflation.
5. Evaluate	Make judgments about the quality, credibility, or relevance of ideas or solutions using criteria and evidence.	Critique the effectiveness of a character's decision. Justify your preferred method of solving this problem. Assess the reliability of this source.
6. Create	Generate new ideas, designs, products, or solutions by combining or reorganizing elements in an original way.	Write a short story using the literary techniques studied. Design an experiment to test a hypothesis. Develop a lesson plan for a given topic.

Important Principle for Teachers

Higher-order questions (Analyze, Evaluate, Create) do not replace lower-order questions — they build on them. A student cannot analyze if they do not first remember and understand.

In most Nigerian classrooms, 80-90% of test items test only Remember and Understand. Challenge yourself to include at least 30% of items at Apply level and above. This produces deeper learning and better prepares students for national examinations that increasingly demand reasoning.

Classroom Activity 2 (15 minutes)

Topic: Any subject you teach.

Step 1: Write one learning objective at each of the six levels of Bloom's Taxonomy for a topic you recently taught.

Step 2: Write one test item (question) that assesses each objective.

Step 3: Exchange with a colleague and identify the level of each other's questions.

Debrief: What patterns did you notice? Were most questions at lower levels?

8. The Affective and Psychomotor Domains

While most classroom testing targets the cognitive domain, teachers must also assess learning in the affective and psychomotor domains — particularly in subjects like Physical Education, Music, Fine Arts, Home Economics, and Technical subjects.

8.1 The Affective Domain (Krathwohl et al., 1964)

Level	Assessment Strategy
Receiving: awareness and willingness to attend	Observation of attention; participation checklists
Responding: active participation and reaction	Participation records; self-report questionnaires
Valuing: attaching importance or commitment to something	Journal entries; attitude surveys; behavioural observation
Organization: prioritizing and resolving conflicts between values	Essays; interviews; personal statements
Characterization: consistent value-based behaviour	Long-term observation; teacher ratings; peer review

8.2 The Psychomotor Domain (Simpson, 1972)

The psychomotor domain covers physical skill development, from initial perception through to originative creation of new motor patterns. It is best assessed through direct observation, performance tasks, and practical examinations using structured rating rubrics.

- Perception (using sensory cues to guide motor activity)
- Set (mental, physical, and emotional readiness to act)
- Guided Response (early-stage copying or trial-and-error)
- Mechanism (performing a skill habitually with confidence)
- Complex Overt Response (fluid, efficient, skilled performance)
- Adaptation (modifying movements for new circumstances)
- Origination (creating new movements to solve a problem)

PART THREE: TEST CONSTRUCTION AND ITEM WRITING

LECTURE THREE: DESIGNING AND WRITING TEST ITEMS

9. The Table of Specifications (TOS)

Before writing a single test item, every teacher should construct a Table of Specifications (TOS) — also called a test blueprint. The TOS is a two-dimensional matrix that maps the content of the test against the cognitive levels being assessed. It is the single most important tool for ensuring content validity.

9.1 How to Construct a TOS

10. List all the major topics covered during the instructional period.
11. Determine the weight (percentage emphasis) for each topic, proportional to the time spent teaching it.
12. Decide which cognitive levels (Remember, Understand, Apply, Analyze, Evaluate, Create) will be tested.
13. Allocate the total number of test items to each topic-level cell, proportionally.
14. Write items to fill each cell before finalizing the test.

Sample TOS — JSS 2 Basic Science (40 Items)

Topic / Level	Remember	Understand	Apply	Analyze	Total	Weight
Cell Biology	4	3	2	1	10	25%
Nutrition & Digestion	3	4	3	2	12	30%
Ecosystems	2	3	2	1	8	20%
Reproduction	3	3	2	2	10	25%
TOTAL	12	13	9	6	40	100%

Notice how the TOS forces balance: no single topic or cognitive level dominates. Without a TOS, teachers often unconsciously overload easy recall questions from the most recently taught topic.

10. Types of Test Items and Rules for Writing Them

10.1 Multiple Choice Questions (MCQs)

MCQs are the most widely used objective item type in Nigerian schools and national examinations. Each MCQ has a stem (question or incomplete statement) and alternatives (options): one correct answer (key) and three or four incorrect options (distractors).

Rules for Writing Excellent MCQs:

15. The stem must clearly state the problem. The student should understand what is being asked without reading the options.
16. Each item must test ONE and only one learning objective.
17. All distractors must be plausible — they should represent common errors, misconceptions, or confusions, not obviously wrong answers.
18. Avoid 'All of the above' and 'None of the above' — they either give away answers or confuse students unfairly.
19. Keep options grammatically consistent with the stem. Each option should be a natural completion of the stem.
20. Randomize the position of the correct answer across items (avoid patterns like always choosing 'C').
21. Avoid negatively worded stems where possible ('Which of the following is NOT...'). If used, bold or capitalize NOT.
22. Keep option lengths roughly equal — students learn to choose the longest option as a heuristic for the correct answer.
23. Avoid irrelevant cues (e.g., 'a/an' at the end of the stem that grammatically matches only one option).
24. Test important learning, not trivial details or trick questions.

Poor MCQ vs. Improved MCQ

POOR: Which of the following is NOT not a type of rock?

A. Igneous B. Sedimentary C. Plastic D. Metamorphic

Problems: Double negative in stem; 'Plastic' is obviously not a rock type, not a plausible distractor.

IMPROVED: Which of the following is classified as an igneous rock?

A. Limestone B. Marble C. Granite D. Sandstone

All options are real rock types (plausible). Only Granite is igneous. Stem is clear and positive.

10.2 True/False Items

True/False items test knowledge of specific facts and the ability to distinguish correct from incorrect statements. They are quick to write and mark but have a critical limitation: a student has a 50% chance of guessing correctly, making them unreliable for high-stakes assessment without modification.

Best Practices:

- Write items that are unambiguously true or false — avoid items that are 'mostly true' or 'true in some contexts.'
- Avoid using absolute terms (always, never, all, none) which signal false answers, and qualified terms (usually, often, sometimes) which signal true answers.
- Keep items brief and focused on one idea.
- Modification strategy: require students to rewrite false statements to make them true. This eliminates guessing and tests deeper understanding.

10.3 Matching Items

Matching items present two columns — premises (left) and responses (right) — that students must match based on a defined relationship. They are efficient for testing associations, definitions, and classifications.

Best Practices:

- Use homogeneous content within each matching set (e.g., all scientists and their discoveries, or all countries and capitals — not a mixture).
- Provide more responses than premises (at least 2-3 extra) to reduce guessing by elimination.
- Limit each set to 8-12 pairs for manageability.
- Arrange premises randomly, not in a sequence that suggests matches.

10.4 Completion / Fill-in-the-Blank Items

Completion items require students to supply a missing word, phrase, number, or symbol. They test recall more rigorously than True/False items because students cannot guess from given options.

Best Practices:

- Write items so only one correct answer is possible. Ambiguous blanks create marking headaches and student frustration.
- Place the blank near the end of the statement so students understand the question context before encountering the gap.

- Do not create a blank for every other word — this reduces the item to a word search rather than a knowledge test.
- Do not lift statements verbatim from the textbook with a word removed — this rewards rote memorization over understanding.

11. Essay and Constructed-Response Items

Essay items require students to compose extended written responses. They are the best tool for assessing higher-order thinking — analysis, evaluation, and creation — but require careful design and structured scoring to achieve acceptable reliability.

11.1 Restricted-Response Essay

A restricted-response essay limits the scope, length, or format of the student's answer. The constraints make scoring more reliable and reduce the influence of writing ability on content scores.

Examples of Restricted-Response Items

'In not more than 200 words, explain the causes of the French Revolution.'

'List and briefly describe three consequences of deforestation.'

'Compare photosynthesis and respiration, focusing on inputs and outputs only.'

11.2 Extended-Response Essay

Extended-response essays allow students considerable freedom in how they organize and express their answers. They are the best format for assessing synthesis, argumentation, creativity, and critical thinking — but are the most challenging to score reliably.

Scoring Methods for Essays:

- Holistic Scoring: The marker reads the entire response and assigns a single score based on overall quality. Fast but less diagnostic.
- Analytical Scoring: The marker assigns separate scores for each dimension (e.g., content accuracy, organization, language use, evidence). More informative, provides specific feedback, higher reliability.

Improving Essay Scoring Reliability — Best Practices

1. Develop a detailed model answer and mark scheme BEFORE administering the test.

2. Use an analytical rubric with clear descriptors at each performance level.
3. Mark all responses to one question before moving to the next (prevents halo effects).
4. Use blind marking where feasible — conceal student names before marking.
5. Have a second marker independently score a sample of responses to check inter-rater reliability.
6. Train co-markers by marking sample scripts together and discussing discrepancies before independent marking begins.

Classroom Activity 3 (20 minutes)

Step 1: Write one MCQ, one True/False item (with rewrite modification), one matching set (5 pairs), and one restricted-response essay question — all on the same topic.

Step 2: Exchange with a partner. Evaluate each other's items using the rules for good item writing.

Step 3: Identify one strength and one improvement suggestion for each item type.

This mirrors the professional item review process used by WAEC and NECO item writers.

PART FOUR: SCORING, STATISTICS, AND INTERPRETATION

LECTURE FOUR: SCORING, STATISTICS, AND ITEM ANALYSIS

12. Developing Marking Schemes and Scoring Rubrics

A marking scheme (or answer key) must be prepared before the test is administered — never after. A post-hoc marking scheme is influenced by student responses and introduces subjectivity.

- Objective tests: Provide an explicit answer key with allocated marks for each item.
- Short-answer items: Specify all acceptable answers with mark allocations.
- Essay items: Provide model answers with an analytical rubric showing marks for each criterion at each level.
- Include instructions for partial credit and common variations (e.g., accept 'H2O' and 'water').

13. Grading Practices

After scoring, raw scores must be interpreted and converted into grades. Different grading systems serve different purposes:

Grading System	Description and Appropriate Use
Percentage-Based	Raw score converted to a percentage. Simple and widely understood. Used across Nigeria's schools (0–100 scale).
Letter Grade (A–F)	Percentages mapped to letter grades with defined cutoffs (e.g., A: 70–100, B: 60–69, C: 50–59, D: 45–49, F: 0–44).
Grade Point Average	Used in higher education. Points assigned per letter grade (A=5, B=4, C=3, D=2, F=0) and averaged across courses.
Pass/Fail (Mastery)	Student either meets the mastery standard or does not. Used in criterion-referenced and competency-based systems.
Standards-Based Grading	Reports performance against specific learning standards: Exceeds / Meets / Approaching / Beginning. Highly diagnostic.

Common Grading Pitfall

Grading on a curve (adjusting everyone's score relative to the class average) is a norm-referenced practice. It means that even if an entire class fails to master the material, some students will still receive 'A' grades simply by outperforming their equally under-prepared peers.

Reserve norm-referencing for selection decisions (JAMB, scholarships). For mastery and certification, use criterion-referenced standards.

14. Essential Statistics for Classroom Assessment

A teacher does not need to be a statistician — but every teacher must understand basic descriptive statistics to interpret and communicate assessment results meaningfully.

14.1 Measures of Central Tendency

Measure	Formula and Interpretation
Mean ($\bar{X}$)	Sum of all scores divided by the number of scores. The arithmetic average. Best for symmetrical distributions. Sensitive to outliers (extreme scores). Most commonly used in education.
Median (Mdn)	Middle score when all scores are arranged in order. Resistant to extreme scores. Best for skewed distributions. If there is an even number of scores, average the two middle values.
Mode (Mo)	The most frequently occurring score. Useful for identifying the most common performance level. A distribution may have one mode (unimodal), two modes (bimodal), or none.

Worked Example

Class scores: 45, 52, 58, 62, 64, 65, 68, 70, 72, 84

Mean = (45+52+58+62+64+65+68+70+72+84) / 10 = 640 / 10 = 64

Median = average of 5th and 6th values = (64+65)/2 = 64.5

Mode = No mode (all scores occur once)

In this case, the mean and median are very close, suggesting a roughly symmetrical distribution — which is typical of a well-constructed classroom test.

14.2 Measures of Variability

Central tendency tells us where the centre of the distribution lies. Variability tells us how spread out the scores are. Both are needed to fully understand a set of results.

Measure	Definition and Use
Range	Highest score minus lowest score. Simple but affected by outliers. Gives a rough sense of spread.
Variance (s^2)	Average of squared deviations from the mean. Basis for standard deviation. Not directly interpretable in the original score unit.
Standard Deviation	Square root of the variance. Most useful measure of spread. Expressed in the same unit as the original scores. A small SD means scores are clustered near the mean; a large SD means scores are widely dispersed.

Interpreting Standard Deviation in the Classroom

If the class mean is 65 and the SD is 5: most students scored between 60 and 70. The class performed consistently.

If the class mean is 65 and the SD is 20: scores ranged widely. Some students scored as low as 25 and as high as 95. This signals great diversity in learning — and likely a need for differentiated instruction.

In a perfectly normal distribution:

68% of students score within mean ± 1 SD

95% of students score within mean ± 2 SD

99.7% of students score within mean ± 3 SD

14.3 Derived Scores and What They Mean

Score Type	Description
Percentile Rank	The percentage of students in the norming group who scored at or below this score. A student at the 75th percentile scored higher than 75% of the comparison group.
z-Score	Standard score = (Raw Score – Mean) / SD. Indicates how many standard deviations above or below the mean a score falls. Mean=0, SD=1. Can be negative.
T-Score	Converts z-scores to a scale with mean=50, SD=10. Eliminates negatives. Formula: T = 50 + 10(z). Easier to communicate to parents.

Score Type	Description
Stanine	Standard nine-point scale. Divides the normal distribution into 9 bands. Mean=5, SD=2. Used in standardized testing for broad performance categories.

15. Item Analysis — Evaluating the Quality of Test Items

After administering a test, item analysis allows teachers to evaluate the quality of individual items and improve future tests. It is a professional practice that separates intentional test design from guesswork.

15.1 The Difficulty Index (p)

The difficulty index (p) measures the proportion of students who answered an item correctly.

Formula and Interpretation

p = (Number of students who answered correctly) / (Total number of students who attempted the item)

Interpretation:
p = 0.90 or above — Item is very easy (may have little discriminating power)
p = 0.30 to 0.70 — Ideal range for most classroom tests
p = 0.10 or below — Item may be too difficult, ambiguous, or poorly written

For maximum discrimination, items around p=0.50 are ideal.

Example: 30 out of 50 students answered item 5 correctly. p = 30/50 = 0.60. This item is of moderate difficulty — well within the ideal range.

15.2 The Discrimination Index (D)

The discrimination index (D) measures how well an item distinguishes between students who performed well on the test overall (upper group) and students who performed poorly (lower group). A good test item should be answered correctly more often by the upper group.

Formula and Interpretation

$D = (U - L) / (n/2)$

Where: U = correct responses in upper 27%; L = correct responses in lower 27%; n = total students

Interpretation:

D = 0.40 and above — Excellent item
D = 0.30 to 0.39 — Good item
D = 0.20 to 0.29 — Marginal — revise if possible
D = 0.19 or below — Poor item — revise or eliminate
Negative D — Dangerous: lower-group students answered correctly MORE than upper group. Discard and investigate.

A negative D suggests a defective item: perhaps it was ambiguous, had two correct answers, or contained a cue that misled high-ability students.

15.3 Distractor Analysis

For MCQ items, distractor analysis examines how many students in the upper and lower groups chose each incorrect option. An effective distractor should attract a meaningful proportion of lower-group students while being avoided by upper-group students. If a distractor is chosen by zero students, it is not functioning — replace it with a more plausible alternative.

Classroom Activity 4 (20 minutes)

Data: You have just administered a 40-item MCQ test to 50 students.

Item 12: 42 students answered correctly. Compute p.

Item 17: In the upper group (top 14 students), 11 answered correctly. In the lower group (bottom 14 students), 4 answered correctly. Compute D and evaluate.

Item 23: p = 0.28. D = -0.15. What would you do with this item?

Discuss: How would you use these three statistics to revise your item bank for next term?

PART FIVE: AUTHENTIC ASSESSMENT AND PROFESSIONAL ETHICS

LECTURE FIVE: AUTHENTIC ASSESSMENT, RUBRICS, AND ETHICS

16. Authentic Assessment

Authentic assessment asks students to apply their knowledge and skills to real-world tasks and contexts — not just to answer decontextualised test questions. It is grounded in the principle that the best evidence of learning is the ability to USE what has been learned.

Grant Wiggins (1990) on Authentic Assessment

Wiggins argued that conventional tests often measure only the ability to perform well on conventional tests — not the ability to use knowledge in meaningful ways. He called for 'tasks that are worthy of the effort to prepare for, and that are educationally valuable in themselves.'

16.1 Performance Assessment

Performance assessment requires students to demonstrate a skill or competency by performing a task. Common in Physical Education, Science, Music, Technical Education, and Language Arts.

- Student delivers a prepared speech in English Language class — assessed on fluency, vocabulary, content, and audience engagement using a rubric.
- Student conducts a chemistry titration experiment — assessed on technique, accuracy, safety, and recording.
- Student demonstrates a traditional dance in Cultural and Creative Arts — assessed on rhythm, coordination, and expression.

What Makes Performance Assessment Authentic?

It mirrors a real task that a practitioner in the field would perform.

It requires judgment, creativity, or decision-making — not merely recall.

It produces a product or performance that can be directly observed and evaluated.

Assessment criteria are known to students in advance.

16.2 Portfolio Assessment

A portfolio is a purposeful, structured collection of student work over time that documents learning, growth, and achievement. Portfolios shift the focus from a single high-stakes moment to a cumulative, evidence-based picture of a learner's journey.

Portfolio Type	Description
Showcase Portfolio	Displays the student's best or most representative work. Used for promotion, parent conferences, or university applications.
Process Portfolio	Includes drafts, revisions, reflections, and feedback at multiple stages of a project. Shows the thinking and improvement process.
Evaluation Portfolio	Contains evidence of meeting specific learning standards or competencies. Used for summative judgment and grading.

- Portfolios must include student reflections — not just collected work. Without metacognitive commentary, a portfolio is merely a folder.
- Assessment criteria for portfolios must be communicated to students at the start of the collection period.
- Digital e-portfolios (using Google Sites, Mahara, or even organized Google Drive folders) are increasingly accessible and powerful.

16.3 Developing Rubrics

A rubric is a scoring guide that describes levels of quality for each criterion being assessed. It is the essential tool for making authentic assessment reliable, transparent, and instructionally useful.

Rubric Type	Characteristics
Holistic Rubric	A single score based on the overall quality of the work. Fast to apply. Less diagnostic feedback. Best for quick summative judgments.
Analytical Rubric	Separate scores for each defined criterion (e.g., content, organization, language, evidence). More time-consuming but far more informative. Provides criterion-specific feedback that guides improvement.
Single-Point Rubric	Only describes the standard for proficiency. Teacher annotates deviations above or below. Encourages growth-focused feedback.

Components of an Effective Analytical Rubric

1. CRITERIA: What dimensions will be assessed? (e.g., for an essay: Content Accuracy, Argument Structure, Use of Evidence, Language Mechanics)
2. PERFORMANCE LEVELS: How many levels? (3-5 is common: Exemplary / Proficient / Developing / Beginning, or 4 / 3 / 2 / 1)
3. DESCRIPTORS: What does each level look and sound like? Be specific and use observable language.
4. POINT VALUES: What are the marks allocated to each level?

Always share rubrics with students BEFORE the task. The rubric is not a secret — it is a learning tool.

16.4 Self-Assessment and Peer Assessment

Self-assessment and peer assessment are powerful tools for developing metacognition, critical thinking, and learner autonomy. They require explicit training and clear criteria to be effective.

- Self-Assessment: Students evaluate their own work against shared criteria. Ask: 'What did I do well? What would I change? What do I still need to learn?'
- Peer Assessment: Students evaluate each other's work using the same rubric. Deepens understanding of quality criteria. Must be conducted in a respectful, structured environment.
- Both should be used formatively — to improve ongoing learning — not primarily as marks toward a grade.

17. Ethics and Fairness in Assessment

Teachers hold significant power in the assessment process. With that power comes professional and ethical responsibility. Assessment ethics is not a peripheral concern — it is central to the integrity of education and the dignity of every student.

17.1 Core Ethical Principles

Principle	Application
Confidentiality	Test content must be kept secure before administration. Student results must only be shared with the student, parents, and authorized educators.
Accuracy	Scores must be recorded and reported accurately.

Principle	Application
	Computational errors must be corrected without delay when discovered.
Fairness	Assessment must not systematically disadvantage students based on gender, ethnicity, disability, language, or socioeconomic background.
No Harm	Assessment should not cause psychological distress, stigmatization, or loss of dignity. Results must be communicated constructively.
Informed Transparency	Students should know the purpose of assessments, how they will be scored, and how results will be used.
Professional Competence	Teachers should only administer, score, and interpret assessments they are trained to use.
Academic Integrity	Examination malpractice undermines the validity of every certificate. Teachers must model and enforce integrity with consistency and without exception.

17.2 Bias in Assessment

Assessment bias occurs when test items or procedures produce systematically different results for subgroups of students — not because of differences in the knowledge or skill being measured, but because of extraneous factors related to group membership.

- Content Bias: Items reference cultural experiences, objects, or events unfamiliar to certain groups (e.g., items assuming all students have seen snow, used a bank, or live in urban areas).
- Language Bias: Complex sentence structures or high reading levels that disadvantage students whose primary language is not English, or who have reading difficulties.
- Gender Bias: Items that assume male or female perspectives, or that consistently feature one gender in professional roles.
- Stereotype Threat: The anxiety that arises when students fear confirming a negative group stereotype. Research shows this can measurably depress performance.

Bias Review Practice

A physics word problem refers to 'John calculating the speed of his sports car.' This may marginalize female students who do not identify with the scenario.

Revised: 'A vehicle travels 240 km in 3 hours at constant speed. Calculate the speed in km/h.' — Neutral, context-free, no bias.

Examination Malpractice — A National Crisis

Examination malpractice in Nigeria — including leakage of question papers, impersonation, and cheating — costs the nation billions in wasted credentials and corrupts the value of every legitimate certificate.

As a teacher, you are the frontline guardian of assessment integrity. This means: secure your test materials, supervise examinations without compromise, refuse inducements, and report irregularities. Professional integrity in assessment is non-negotiable.

18. Technology in Assessment

Digital tools are transforming the possibilities for assessment design, delivery, feedback, and analysis. Every 21st-century teacher should be aware of and developing competence with the following:

- Computer-Based Testing (CBT): Enables adaptive testing, instant scoring, large-scale administration, and multimedia items. JAMB UTME has used CBT since 2013.
- Online Quiz Tools: Google Forms, Kahoot, Quizizz, and Socrative allow rapid formative quizzes with immediate automated feedback and data.
- Learning Management Systems (LMS): Platforms like Google Classroom, Moodle, and Canvas host assignments, grade books, rubrics, and feedback in one integrated environment.
- Spreadsheet Analysis: Microsoft Excel and Google Sheets can compute means, standard deviations, and item analysis data quickly.
- Plagiarism Detection: Turnitin, Grammarly, and similar tools verify academic integrity in written work.
- E-Portfolios: Platforms like Seesaw, Google Sites, and Mahara allow students to curate digital evidence of learning.

Caution on Technology

Technology is a tool — it amplifies good assessment design and also amplifies poor design. A poorly worded MCQ delivered by CBT is still a poorly worded MCQ. Master the principles of assessment before automating the process.

Also consider equity: not all students have equal access to technology. Ensure technology-mediated assessments do not inadvertently disadvantage students without reliable internet or devices.

PART SIX: CONTINUOUS ASSESSMENT AND NATIONAL CONTEXT

LECTURE SIX: CONTINUOUS ASSESSMENT AND NATIONAL EXAMINATIONS

19. Continuous Assessment (CA) in Nigeria

The Federal Ministry of Education of Nigeria introduced Continuous Assessment as a compulsory component of the national evaluation framework. CA replaces the singular reliance on end-of-term examinations with a systematic, ongoing collection of evidence about student learning throughout the academic term.

19.1 Definition and Rationale

Continuous Assessment is a systematic, ongoing method of evaluating student progress during the instructional process using multiple instruments and methods across cognitive, affective, and psychomotor domains. Its introduction was a response to the well-documented limitations of single-point, high-stakes examinations.

- Single end-of-term exams are unreliable — a student may have an unusually good or bad day.
- They test only a narrow snapshot of learning, not the full breadth of objectives.
- They do not measure growth, process, or affective development.
- They encourage last-minute cramming rather than sustained deep learning.

19.2 Components of Continuous Assessment

CA Component	Description and Weighting Guidance
Class Tests	Periodic short tests on recently taught content (2-3 per term minimum). Typically 30-50% of CA score.
Assignments & Projects	Homework, class exercises, and extended project tasks. Assesses application and effort. Typically 20-30% of CA score.
Class Participation	Observation of engagement, questioning, and contribution in class discussions. Requires structured observation rubric.
Practical/Laboratory Work	For Science, Technical, and Vocational subjects. Assesses psychomotor skills and procedural understanding.

CA Component	Description and Weighting Guidance
Attitude and Conduct	Affective domain indicators: punctuality, cooperation, responsibility. Observed over time using rating scales.

19.3 CA Weighting in the Final Grade

In most Nigerian secondary schools, the CA contributes 30-40% to the final subject grade, with the End-of-Term or End-of-Year examination contributing the remaining 60-70%. In primary schools, CA may carry up to 60% weight under the UBE policy framework.

Professional Responsibility

CA records must be maintained accurately, updated consistently, and kept securely. Fabricating or manipulating CA scores is a serious professional and legal violation.

CA data should also be used formatively — not just entered into a grade book, but used to identify students who need support, topics that need reteaching, and parents who need to be informed.

20. Nigeria's National Examination Framework

Nigerian teachers must understand the structure and demands of the national examinations for which they prepare students. These examinations are the primary accountability mechanisms for the secondary school curriculum.

Examination	Key Information for Teachers
WAEC SSCE (West African Examinations Council)	End of SS3. Multi-country (Nigeria, Ghana, Sierra Leone, Gambia, Liberia). Includes objective, theory, and practical papers. Grades: A1 through F9. Credit (A1-C6) in five subjects including English and Mathematics is required for most university admissions.
NECO SSCE (National Examinations Council)	Nigerian national alternative to WAEC. Administered by NECO. Similar format and grading to WAEC. Increasingly accepted for university admission.
JAMB UTME (Joint Admissions and Matriculation Board)	Gateway to all Nigerian universities and polytechnics. Computer-Based Test (CBT) since 2013. Four subjects: English plus three others matching intended course of study. Scores combined with O-level results for university admission.
BECE (Basic Education Certificate Examination)	End of JSS3. Conducted by State Examination Boards. Validates completion of the 9-year basic education. Required for entry into senior secondary school.

Examination	Key Information for Teachers
NECO NBCE (National Business Certificate Exam)	For technical colleges. Certifies technical and vocational competencies.

20.1 How Teachers Should Use National Examination Syllabuses

- Download and study the official syllabus for every subject you teach from WAEC (www.waecnigeria.org) and NECO (www.neco.gov.ng).
- Align your term plan to the syllabus objectives — not just the textbook.
- Review past questions (going back 10 years) to understand item format preferences, recurrent topics, and the cognitive level of questions.
- Administer mock examinations under timed conditions with authentic past-question formats.
- Analyze mock results using item analysis to identify topics that need more classroom attention.
- Teach students explicitly how to read and respond to different item formats — many students lose marks not from lack of knowledge but from misreading questions.

LECTURE SUMMARY AND KEY TAKEAWAYS

This lecture has provided a comprehensive tour of measurement and evaluation in teaching. Here is a consolidated summary of the most important professional insights:

Theme	Core Professional Takeaway
Definitions	Measurement, assessment, evaluation, and testing are distinct. Know the difference and use each concept precisely.
Purposes	Assessment serves instructional, diagnostic, placement, certification, motivational, and accountability purposes. Design with purpose.
Principles	Every assessment must be valid, reliable, objective, comprehensive, practicable, and fair. These are non-negotiable professional standards.
Bloom's Taxonomy	Teach and test across all six cognitive levels. Overreliance on recall-level questions impoverishes assessment and stifles deep learning.
Item Writing	Good items are written, reviewed, and revised by professionals. Learn and apply the rules for MCQs, True/False, matching, and essays.

Theme	Core Professional Takeaway
Table of Specifications	Never design a test without a TOS. It is your blueprint for content validity and balance.
Item Analysis	Use difficulty and discrimination indices to evaluate and improve your item bank after every test.
Statistics	Mean, median, mode, and standard deviation are practical tools for understanding class performance. Know how to compute and interpret them.
Authentic Assessment	Design tasks that require real application of knowledge. Use rubrics consistently. Involve students in self and peer assessment.
Ethics	Assessment integrity is a professional obligation. Fairness, accuracy, confidentiality, and zero tolerance for malpractice are non-negotiable.
National Context	Know your WAEC and NECO syllabuses. Use past questions. Prepare students for the format and demands of national examinations.

FINAL REFLECTION ACTIVITY

Individual Reflection (10 minutes) — To Be Submitted

On a sheet of paper, respond to the following three prompts:

1. INSIGHT: What is the single most important thing you learned in this lecture that you did not know before, or that clarified a previous misunderstanding?

2. APPLICATION: Identify one specific change you will make to your assessment practice in the next two weeks as a result of this lecture. Be as specific as possible.

3. QUESTION: What is one question this lecture raised for you that you would like to explore further?

A Final Word for Teachers

"The aim of education is not to pass tests. The aim of education is to develop human beings who can think, create, and contribute to their communities. Assessment, at its best, serves that aim — by showing us where we are, and pointing us toward where we need to go."

Use your assessment tools not as weapons of judgment, but as mirrors of growth — for your students and for yourself as a teacher.

RECOMMENDED FURTHER READING

- Bloom, B. S. et al. (1956). Taxonomy of Educational Objectives: Handbook I — Cognitive Domain. Longmans.
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- Gronlund, N. E. & Waugh, C. K. (2009). Assessment of Student Achievement (9th ed.). Pearson.
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End of Lecture — Measurements and Evaluation

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